

Collin Longoria

C++ Engine & Graphics Programmer focused on high-performance real-time systems and GPU optimization

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EDUCATION

DigiPen Institute of Technology

Redmond, WA

Bachelor of Science in Computer Science in Real-Time Interactive Simulation, Minor in Math Aug. 2022 – May 2026

3.6 GPA

RELEVANT PROJECTS

Cisalpine Engine, Software Engineer | C++, OpenGL, GLFW, Dear ImGui

Jan. 2026 – Present

- Designed and implemented a real-time 2D Radiance Cascades global illumination system with 4-level cascade hierarchy, achieving full-scene indirect lighting at 0.8 ms per frame on RTX-class hardware, enabling dynamic light propagation from emissive elements.
- Built a GPU-driven rendering pipeline with deferred-style G-buffer extraction (color, normals, emission/opacity), multi-pass cascade merging, and final resolve to lightmap, supporting 10+ debug visualization modes for rapid iteration.
- Optimized compute shader dispatch using an indirect draw buffer and chunk-based spatial activation, reducing GPU workload by processing only active regions and achieving a 50% frame time reduction; applied data-oriented design and large-scale workload optimization principles.

Blok (Academic Project), Graphics Programmer | C++, Vulkan, CUDA, GLSL

Aug. 2025 – Present

- Developed a dual-backend voxelization and path-tracing engine with both Vulkan-compute and CUDA-accelerated OpenGL pipelines, enabling flexible GPU performance benchmarking across APIs.
- Engineered a modular Vulkan renderer supporting data-driven pipelines and dynamic descriptor allocation, resulting in a more adaptable and extensible graphics framework.
- Designed a numerically robust chunk-grid DDA traversal with hierarchical Sparse Voxel Octree (SVO) acceleration and distance-based LOD, reducing unnecessary ray work in large voxel scenes and reducing frame time by over 2 ms.

Project Mirrors (Academic Project), Gameplay Programmer | Unreal Engine 5

Aug. 2024 – April 2025

- Developed a modular puzzle framework in C++ allowing designers to link interactive components without code, reducing technical bottlenecks in level design.
- Collaborated with technical artists to design high-performance custom shader effects and reusable Blueprint components, improving gameplay clarity and achieving project goal of 60 frames per second across all target platforms.

Elementokens (Academic Project), Engine Programmer | C++, OpenGL, FMOD

Aug. 2023 – April 2024

- Led engine-side development of rendering and gameplay logic, creating reusable modules for turn-based movement, combat, and dynamic map rendering to streamline feature integration.
- Designed a data-driven content pipeline that enabled game designers to prototype units, maps, and rulesets without engine changes, accelerating iteration cycles by roughly 150%.

EXPERIENCE

Assistant Teacher

June 2025 – July 2025

DigiPen WANIC Program

Redmond, WA

- Mentored high school students in programming fundamentals, data structures, and basic game programming in JavaScript.
- Debugged student code in real-time, identifying logic errors and teaching debugging strategies.
- Developed supplemental coding exercises and mini-projects to reinforce classroom material, including interactive projects using P5.js.
- Collaborated with the lead instructor to adapt lesson content based on student feedback and performance.

TECHNICAL SKILLS

Languages: C++, C, CUDA, GLSL, HLSL, C#, Python, JavaScript

Engines & APIs: Unreal Engine 5, Godot, OpenGL, Vulkan, DirectX 12, WebGPU

Developer Tools: Git, Perforce, Visual Studio, VS Code, CLion, CMake, Renderdoc, NVIDIA NSight

Libraries: glslang, FMOD, Assimp, stb, GLM, SDL, GLFW, Dear ImGui